HR Workforce & Employee Attrition Analysis

Employee Workforce Insights, Attrition Trends & Key HR Drivers

Employees Attrition by Gender



Job Role

All

Department

- Human Resources
- Research & Development
- Sales

Over Time

- No
- Yes

Gender

Female

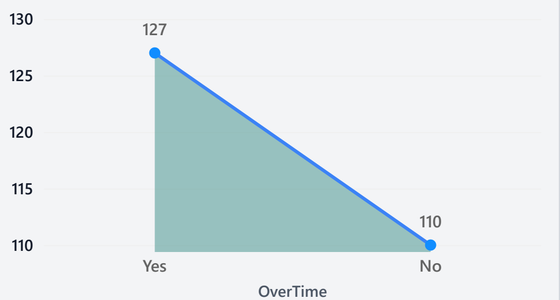
Male

Business Travel

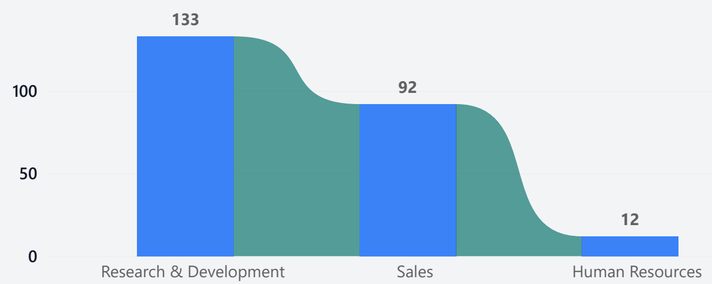
- Non-Travel
- Travel_Frequently
- Travel_Rarely

Average Monthly Income	Active_Employees	Average Employee Age	Total Employees	Employees Attrition	Employee Attrition Rate
6.50K	1233	36.92	1470	237	16.12

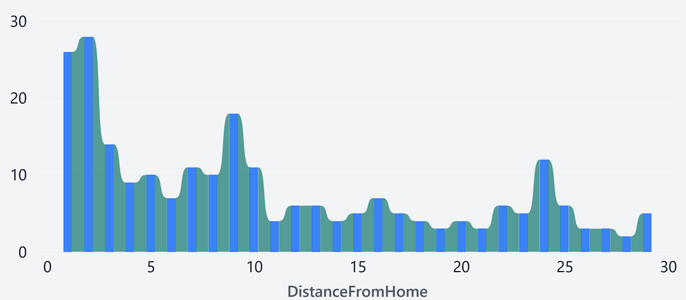
Employees Attrition by Over Time



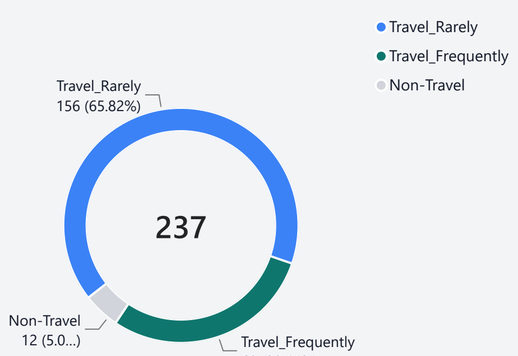
Employees Attrition by Department



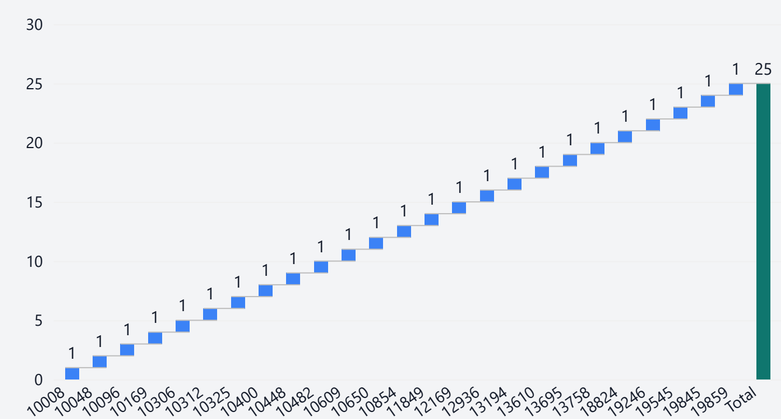
Employees Attrition by Distance From Home



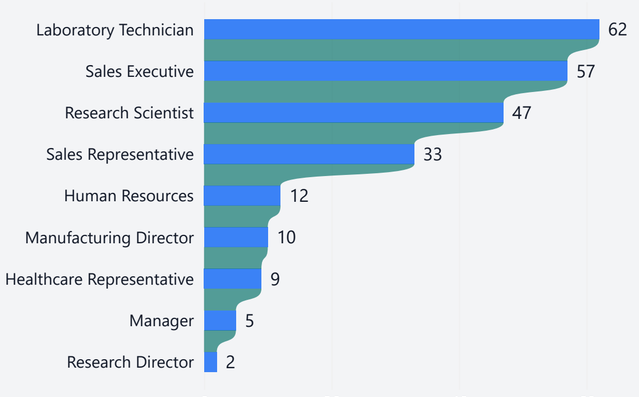
Employees Attrition by Business Travel



Employees Attrition having Monthly Income More than 10000



Employees Attrition by Job Role



HR WORKFORCE & EMPLOYEE ATTRITION ANALYSIS
FINAL REPORT

1. Project Objective

- The objective of this analysis is to examine employee workforce and attrition patterns and identify employee and workplace factors associated with higher employee attrition.
- The analysis focuses on understanding the overall workforce structure and examining attrition across factors such as overtime, monthly income, distance from home, department, business travel, and gender.
- The analysis was performed using Python for data analysis and visualization, SQL for data querying, and Power BI for interactive dashboard development.

2. Executive Summary

- This report presents an analysis of employee workforce and attrition patterns based on a dataset containing 1,470 employees.
- The analysis identified 237 employee attrition cases, resulting in an overall attrition rate of 16.12%. Several notable patterns were observed across employee workload, compensation, workplace location, department, business travel, and gender.
- The findings indicate that employee attrition is not evenly distributed across the workforce. Certain employee groups show considerably higher numbers of attrition cases, highlighting areas that may require further investigation from an HR and employee-retention perspective.

3. Workforce Overview & Key Performance Indicators

Key Performance Indicator	Value
Total Employees	1,470
Active Employees	1233
Employees Attrition	237
Attrition Rate	16.12%
Average Monthly Income	6.5k
Average Employee Age	36.92 years

- The organization has a total workforce of 1,470 employees, of which 1,233 employees remain active and 237 employees have exited the organization. The overall employee attrition rate is 16.12%.
- The average monthly income of employees is 6.5K, while the average employee age is 36.92 years.

4. Key Findings

Finding 1 — Overtime and Employee Attrition

- Employees working overtime recorded **127 attrition cases**, while employees who did not work overtime recorded **110 attrition cases**.
- This indicates that employees working overtime have a higher observed number of attrition cases, suggesting that workload and overtime patterns may be important areas for further HR investigation.

Finding 2 — Monthly Income and Employee Attrition

- Employees with a monthly income of **10K or above recorded 25 attrition cases**, whereas employees earning **below 10K recorded 212 attrition cases**.
- The substantial difference between these groups highlights a strong association between income level and observed employee attrition within the analyzed workforce.

Finding 3 — Distance from Home and Employee Attrition

- The analysis shows that employees living at a **shorter distance from the workplace recorded higher attrition**, while employees living at a **longer distance recorded comparatively lower attrition**.

Finding 4 — Research & Development Department Has Higher Attrition

- The **Research & Development (R&D) department recorded the highest number of employee attrition cases** among the analyzed departments.

- This makes R&D an important area for further investigation, particularly to understand whether specific job roles, workload patterns, compensation levels, or employee experience factors are associated with the higher attrition observed in this department.

Finding 5 — Business Travel and Employee Attrition

- Employees who **travel rarely for business recorded a higher number of attrition cases** compared with employees in the other business-travel categories.
- This indicates that business-travel frequency is associated with differences in observed employee attrition and may be worth examining alongside other employee and job-related factors.

Finding 6 — Gender-wise Employee Attrition

- **Male employees recorded 150 attrition cases**, while female employees recorded **87 attrition cases**.
- The analysis therefore shows a higher number of observed attrition cases among male employees in this workforce.

5. Key Business Implications

The analysis highlights several areas that HR management could examine more closely when developing employee-retention strategies.

I. Workload and Overtime

- The higher number of attrition cases among employees working overtime suggests that workload and overtime patterns may deserve closer attention, particularly in teams where overtime is consistently high.

II. Compensation

- The large difference in attrition between lower-income and higher-income employee groups indicates that compensation should be considered as an important area when examining employee retention.

III. **Department-level Retention**

- The higher attrition observed in the Research & Development department suggests that department-specific analysis may be useful for identifying the underlying factors associated with employee turnover.

IV. **Employee Characteristics**

- Differences in attrition across gender, business-travel frequency, and distance from home demonstrate that employee attrition can vary considerably across different workforce segments.

6. Analytical Limitations

- The findings in this report represent patterns and associations observed within the analyzed dataset. They should not be interpreted as proof that a particular factor directly causes employee attrition. Additional organizational information and deeper analysis would be required to establish causal relationships.

7. Tools & Technologies

I. **Python:** Data preprocessing, cleaning, analysis, and visualization.
II. **SQL:** Data querying and analytical investigation.
III. **Power BI:** Interactive dashboard development and visualization.
IV. **DAX:** HR metrics and KPI calculations.
V. **Excel:** Data handling and supporting analysis.

8. Conclusion

The HR workforce and attrition analysis provides an overall view of employee retention patterns across the organization.

With **1,470 employees and an attrition rate of 16.12%**, the analysis identifies several workforce segments with notable differences in observed attrition. Overtime, monthly income, distance from home, department, business travel, and gender all show meaningful differences in attrition patterns within the dataset.

The findings can help HR teams identify areas requiring deeper investigation and support more targeted employee-retention strategies. Combining these findings with additional employee satisfaction, job-role, compensation, and workload information could provide a more comprehensive understanding of employee turnover.

Overall, the analysis demonstrates how HR data can be transformed into actionable workforce insights and used to support data-driven decision-making.